INCH-POUND

MIL-PRF-1/390E
3 August 1998
SUPERSEDING
MIL-E-1/390D
8 November 1971

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY
TYPES 3RP1 AND 3RP1A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Electrostatic deflection and focus (3RP1 has a curved face and the 3RP1A has a flat face).

PIN CONNECTIONS AND DIMENSIONS: See figure 1.

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Light output	Rg	Zd	Alt
Unit:	V	V dc	v	V dc	V dc	fL	Megohms	Megohms	ft
Maximum:	6.9	0, -200	550	1,100	2,750	- - -	1.5	1.0	10,000
Minimum:	5.7	- - -	- - -	- - -	500 note 1	4.0	- - -	- - -	- - -
Test conditions:	6.3	Adjust	- - -	Focus	1,000	- - -	- - -	- - -	- - -

GENERAL:

Qualification - Required.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE 1. Testing and Inspection.

Inspection	Method	Conditions	Symbol	Limits Min	Limits Max	Unit
Qualification inspection						
Base material insulating quality	1216	Zone 5 (minimum)	- - -	- - -	- - -	- - -
Pressure (implosion)	1141		- - -	- - -	- - -	- - -
Vibration	5111		Width	- - -	1.0	mm
Direct interelectrode capacitance	1331	g1 to all D1 to D2 D3 to D4 D1 to all except D2 D2 to all except D1 D3 to all except D4 D4 to all except D3	Cg1 C1D2 C3D4 CD1 CD2 CD3 CD4	- -	11.0 4.0 4.0 15.0 11.0 10.0 11.0	pF pF pF pF pF pF pF
Neck and bulb alignment (electrostatic types)	5101		Dia	- - -	1.63	inch
Deflection-factor uniformity	5248		- - -	- - -	- - -	- - -
Conformance inspection, part1						
Bulb, screen and faceplate quality	5106		- - -	- - -	- - -	- - -
Voltage breakdown	5201		- - -	- - -	- - -	- - -
Voltage breakdown (electrostatic types)	5201		- - -	- - -	- - -	- - -
Gas “cross”	5206	Light = 4 fL note 2	- - -	- - -	- - -	- - -
Light output	5221	See note 2	Light	4.0	- - -	fL
Modulation	5223	Light = 4 fL	ΔEc1	- - -	48	V dc
Spot position (electrostatic deflection)	5231		- - -	- - -	15.0	mm
Spot displacement (leakage)	5231		Displ	- - -	7.0	mm
Grid cutoff voltage	5241		Ec1	- - -	-67.5	V dc
Grid No. 1 leakage current	5251		- - -	- - -	- - -	- - -
Anode No. 1 leakage current	5251		- - -	- - -	- - -	- - -
Anode No. 2 leakage current	5251		- - -	- - -	- - -	- - -
Conformance inspection, part 2						
Secureness of base, cap or insert	1101		- - -	- - -	- - -	- - -
Base pin solder depth	1111		- - -	- - -	- - -	- - -
Heater current	1301		If	540	660	mA
Electrode current (anode No. 1)	5201	Ec1 = 0	Ib1	-15	10	µA dc
Electrode current (cathode)	5201	Light = 4 fL	Ik	- - -	1,000	µA dc

See notes at end of table I.

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TABLE 1. Testing and inspection - Continued.

Inspection	Method	Conditions	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 2 -Continued.						
Neck and base alignment (electrostatic types)	5101		- - -	- - -	- - -	- - -
Base and trace alignment	5101	+3D4; pin no. 1	- - -	- - -	- - -	- - -
Angle between traces	5101		- - -	- - -	- - -	- - -
Stray light emission (conventional types)	5216	Eb2 = 2,750 V dc	- - -	- - -	- - -	- - -
Line width A (electrostatic deflection)	5226	Light = 4 fL	Width	- - -	0.65	mm
Line width B (electrostatic deflection)	5226	Light = 4 fL	Width	- - -	0.65	mm
Focusing voltage at cutoff	5246	Light = 4 fL	Eb1	165	310	V dc
Focusing voltage at modulation condition	5246		Eb1	165	310	V dc
Deflection factor	5248	1D2	DF	73	99	V dc/inch
Deflection factor	5248	3D4	DF	52	70	V dc/inch
Heater-cathode leakage current	5251		- - -	- - -	- - -	- - -
Permanence of marking	1105		- - -	- - -	- - -	- - -
Conformance inspection, part 3						
Life test	- - -	Group C; light = 4 fL; Eb2 = 2,750 V dc; T= 500 hours (min)	- - -	- - -	- - -	- - -
Life-test end points:	- - -	Light = 3.4 fL	- - -	- - -	- - -	- - -
Line width A			Width	- - -	0.65	mm
Line width B			Width	- - -	0.65	mm
Modulation			ΔEc1	- - -	48	V dc
Grid No. 1 leakage current			- - -	- - -	- - -	- - -
Anode No. 1 leakage current			- - -	- - -	- - -	- - -
Anode No. 2 leakage current			- - -	- - -	- - -	- - -
Heater-cathode leakage current			- - -	- - -	- - -	- - -
Stray light emission			- - -	- - -	- - -	- - -

NOTES:

1. This value is recommended only for low velocity deflecting and low ambient light levels.
2. This test to be performed at the conclusion of the holding period.

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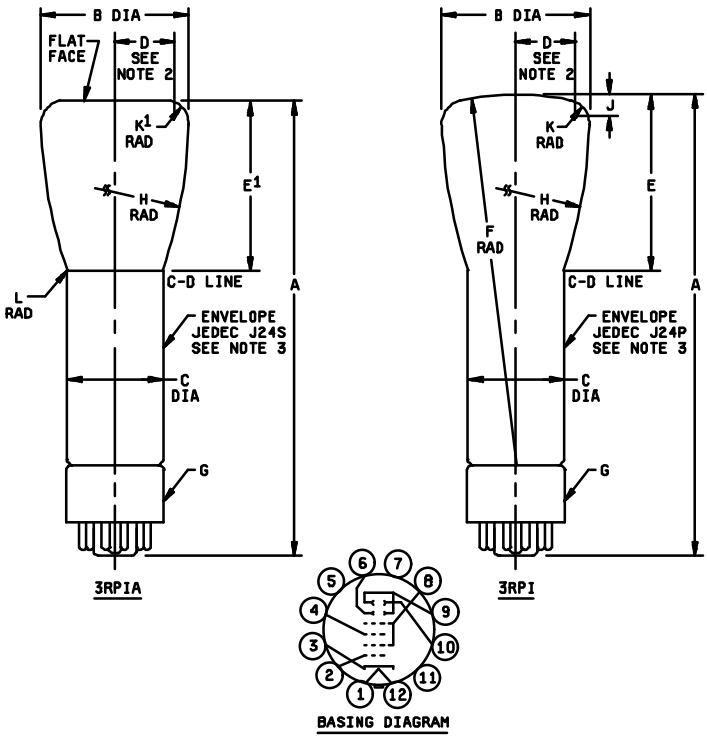

Dimensions are in inches with metric equivalents (mm) in parentheses		
Ltr	Minimum	Maximum
Qualification inspection		
G	Base: B12-43 or B10-75	
Conformance inspection, part 2		
A	8.875 (225.43)	9.375 (238.13)
B	2.880 (73.15)	3.140 (79.76)
C	- - -	1.470 (37.34)
D	1.370 (34.80)	- - -
Reference dimensions (see note 1)		
E	3.875 (98.43)	
E^1	3.563 (90.50)	
F	8.000 (203.20)	
H	17.000 (431.80)	
J	0.350 (8.89)	
K	0.250 (6.35)	
K^1	0.125 (3.18)	
L	0.500 (12.70)	

NOTES:

1. These dimensions are for information only and are not required for inspection purposes.
2. Useful screen radius.
3. Clear or filter glass.

FIGURE 1. Outline drawing of electron tube types 3RP1 and 3RP1A.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS
Air Force - 11, 17, 99

Preparing activity:
DLA - CC

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